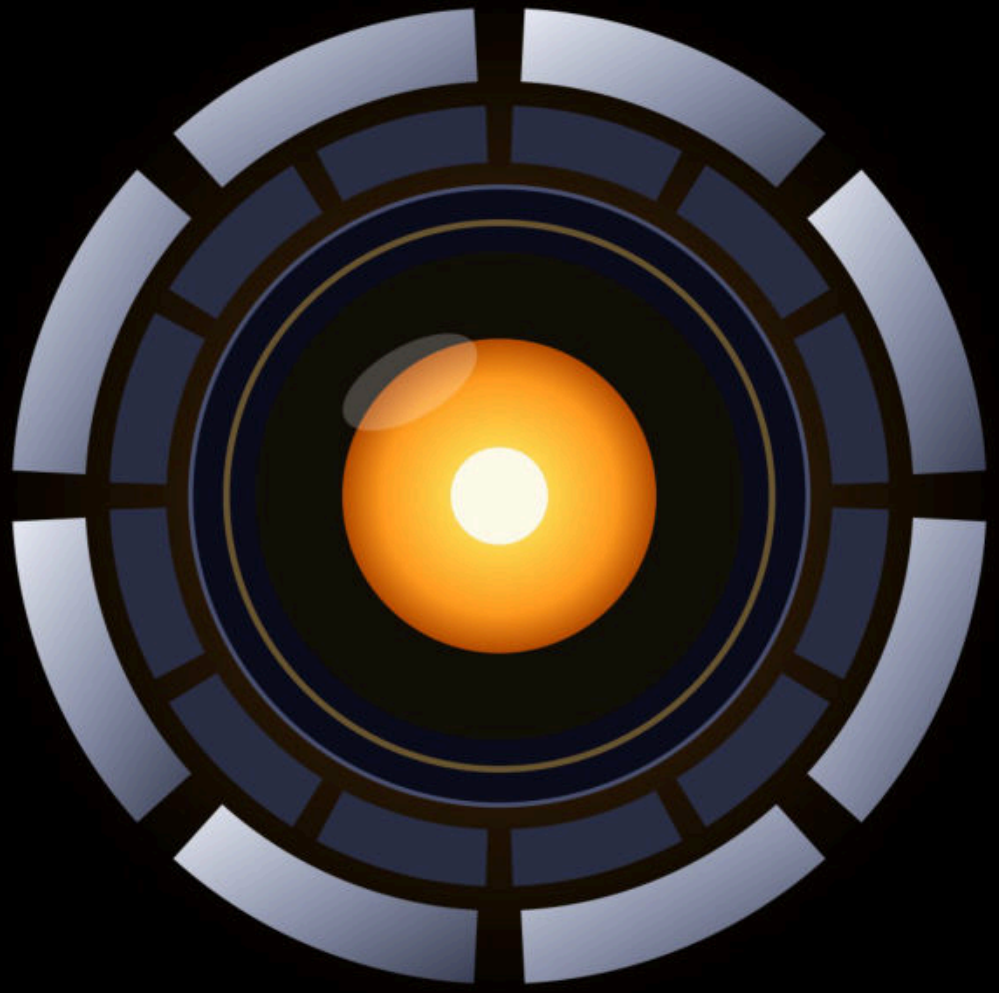
AI 4 DEV · Brno · 12 Nov 2026

GLaDOS

my personal pi agent

A local AI agent on a gaming PC. Discord, voice, vision, images and full computer control.



Lukas Chylik · github.com/depthark

Test plan

Six test chambers

01 GLaDOS and the stack

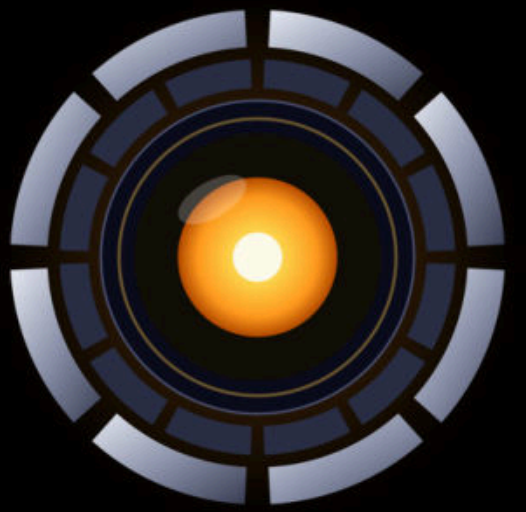
02 Inference: 35B in 12 GB

03 Qwen 3.6 and pi

04 Discord and the decision model

05 Voice, vision, images, browser, desktop

06 What broke and what it costs



6 live demos
“The tests are safe. Mostly.” — GLaDOS

One gaming PC.
Six AI models.
Zero API bills.

I run it from Discord. Phone or PC, 24/7.

Architecture

Four layers, one PC

Discord
text · voice · images

Bridge
Node + discord.js

pi agent
tools · skills

llama.cpp
router :8080

Models

Qwen 3.6 35B · d1-2B · LFM3-VL
Qwen-Image 2.1 · Whistle

Hands

bash · agent-browser
agent-desktop · GitHub

In the background

AI brief · CSS news · free games
cinema · watchdog · WattLlama

Terminal

Herdr: mission control for agents

Many agents at once

panes, tabs, workspaces: pi, Claude Code, OpenCode

Knows who is waiting for me

working · done · blocked on an approval, plus a notification

Nothing gets lost

survives a closed window; a restart resumes the session

Starts with the PC

Windows Terminal → Herdr → pi, zero clicks

The pi extension

```
// herdr-agent-state.ts
pi.on("session_start")
  // reports the session ID
pi.on("agent_start")
  // → working
pi.on("agent_settled")
  // → idle / done
ui.custom({ blocked })
  // → blocked on an approval
```

Outside Herdr it does nothing, so GLaDOS on Discord is unaffected.

Warp → Herdr · open source, Apache 2.0 · since 3 Nov 2026

Inventory

Every model, one PC

MODEL	ROLE	DISK	VRAM	RAM	WHEN IT RUNS
Qwen 3.6 35B-A3B (MTP)	brain: chat, tools, news	22.7 GB	10.8 GB	~19 GB*	always · GPU +8 cores
LiquidAI d1-2B	decides: yes/no, choice, score	1.1 GB	0	1.0 GB	always · CPU, 1–4 s
LiquidAI LFM3-VL-2B	vision: describes chat images	1.3 GB	0	1.9 GB	always · CPU, ~20 s
Qwen-Image 2.1 + Turbo	draws and edits images	8.4 GB	~6 GB	~15 GB	while drawing · 20–80 s
Qwen3-VL-8B + VAE	reads the prompt, decodes pixels	6.9 GB	shared	shared	with Qwen-Image
Cactus Whistle	voice note to text (English)	17 MB	0	tiny	per voice note · 0.3 s

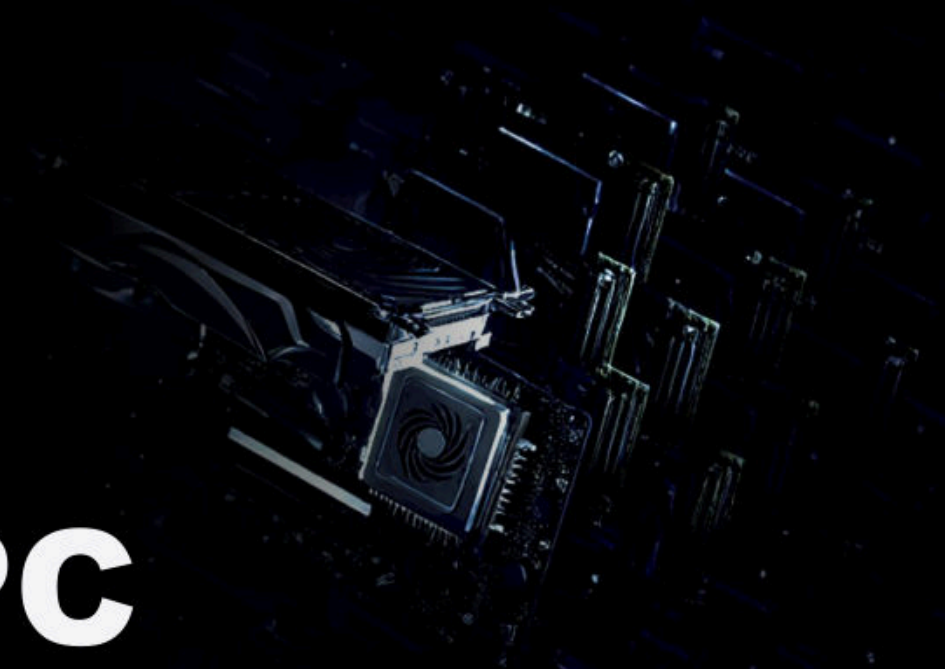
* experts in RAM are a memory-mapped file (mmap): 18.7 GB from disk, the rest is buffers

~40 GB on disk **11.5 / 12 GB** VRAM at idle **~26 / 48 GB** RAM, whole PC

02

Inference on a gaming PC

How a 35B model fits into a 12 GB GPU.



Hardware

The lab = a gaming PC

RTX 4070

12 GB VRAM

i7-9700K

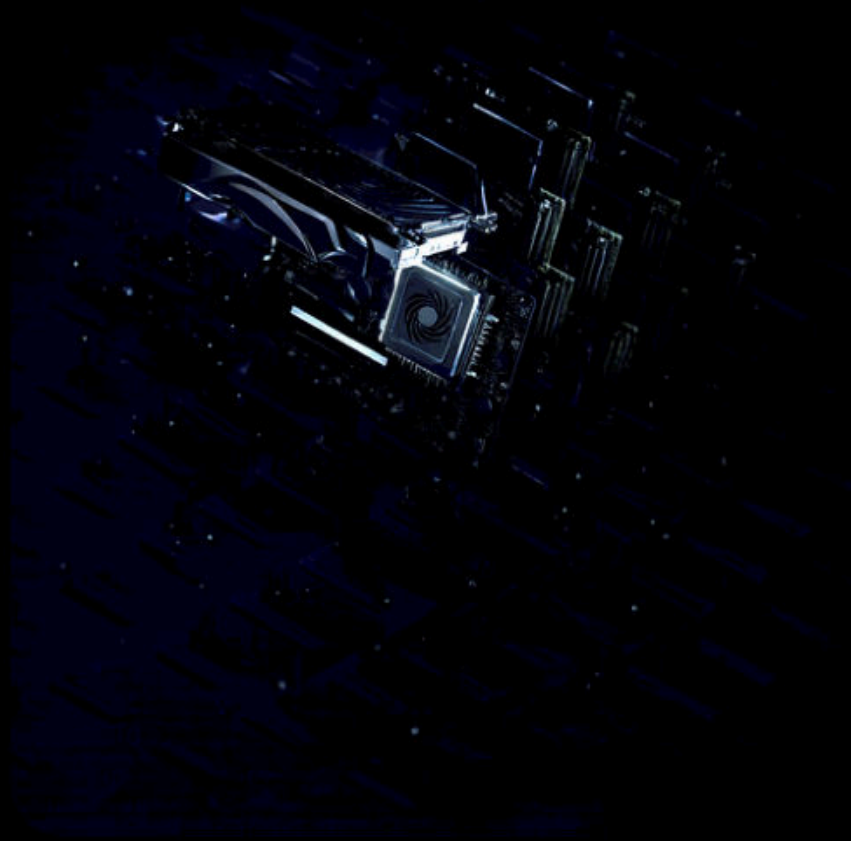
8 cores, 9th gen

48 GB

DDR4-2400

PCIe 3.0

x16 · this will matter



“It is not a data center. It is a gaming PC that no longer plays games.” — GLaDOS

Finding a brain that fits 12 GB

UNTIL JUN 2026

Qwen2.5-14B

dense

whole model on the GPU

- weak answers, previous generation
- fine for chat, lost in tool chains

JUN – SEP 2026

Qwen3-14B

~25 tok/s

dense · 11.9 GB VRAM · 32k context

- thought ~300 tokens per reply → reasoning-budget = 0
- called a real search result "speculation" → SOUL.md: tool output is ground truth
- fumbled agent-browser (>280 s) → wrapped as a tool: 0.4–1.1 s

5 SEP 2026

Qwen3.6-35B-A3B

34.5 tok/s

MoE · 35B total, 3B active per token

- a dense 27B would not fit one 12 GB card
- I predicted 12–18 tok/s; only 2 KV heads made it 34.5
- first replies came back empty: it ignores reasoning-budget → reasoning = off

Mixture of Experts

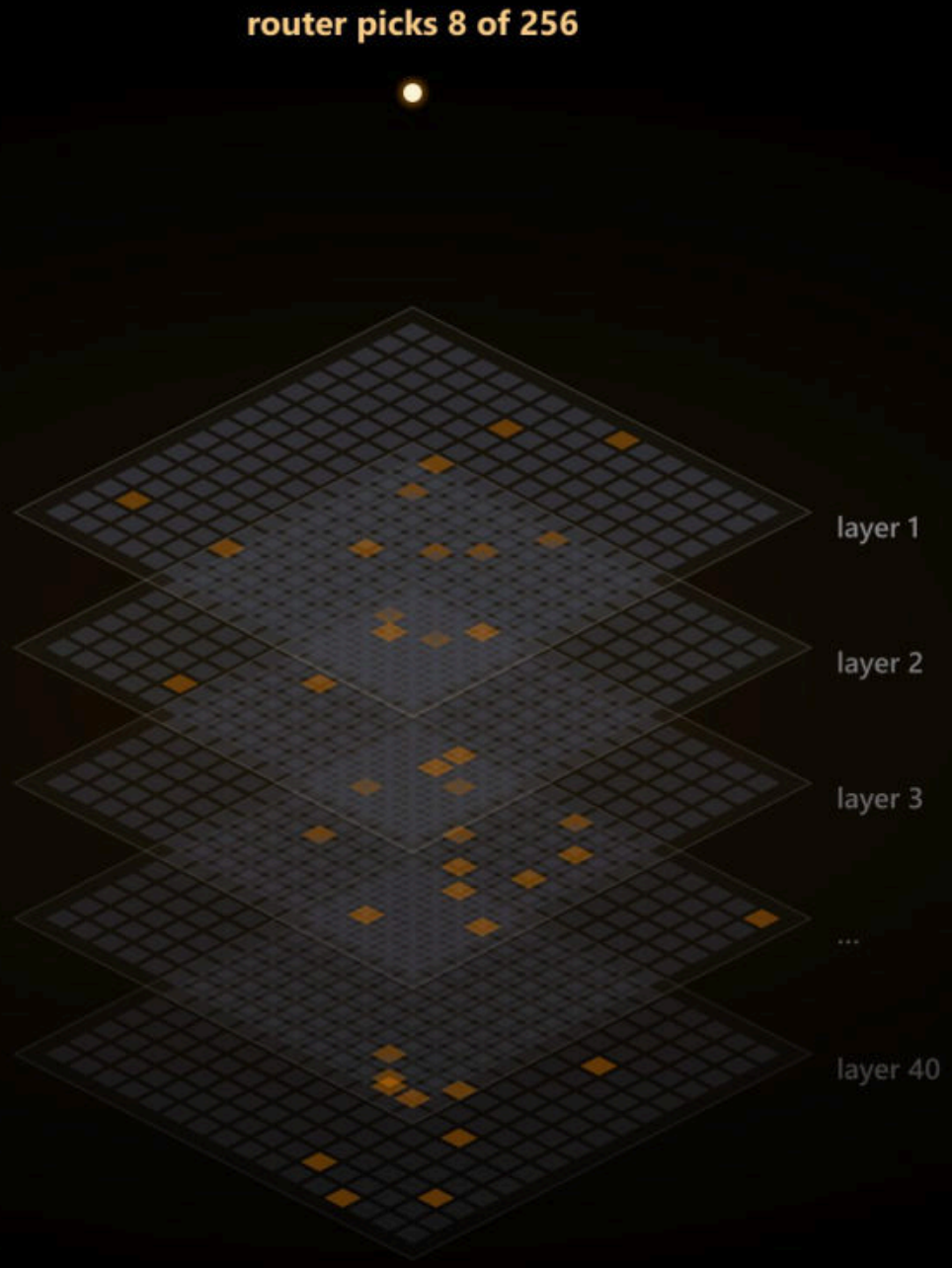
35 billion parameters. 3 billion per token.

8 / 256

experts per token in each of the 40 layers

21.1 GiB

the whole model, Q4_K_M, 5.1 bits per weight



What runs where

Attention on the GPU. Experts in RAM.

Measured: llama.cpp buffer report, 2 Nov 2026

Then making it fast

5 SEP

n-cpu-moe = 24

34.5 tok/s

experts of 24 layers in RAM

- 20 is a cliff: VRAM full, the driver pages, 14.9 tok/s

27 SEP

MTP variant + 64k

+44 %

code 36.5 → 52.4 tok/s

- same weights plus a 0.5 GB draft head
- 3 drafts gave nothing more

27 SEP

ubatch = 2048

2× prefill

338 → 656 tok/s at 19k

- experts cross PCIe once per batch
- ubatch 4096 collapses to 154

16 OCT

moe-cache = 6400

56 tok/s

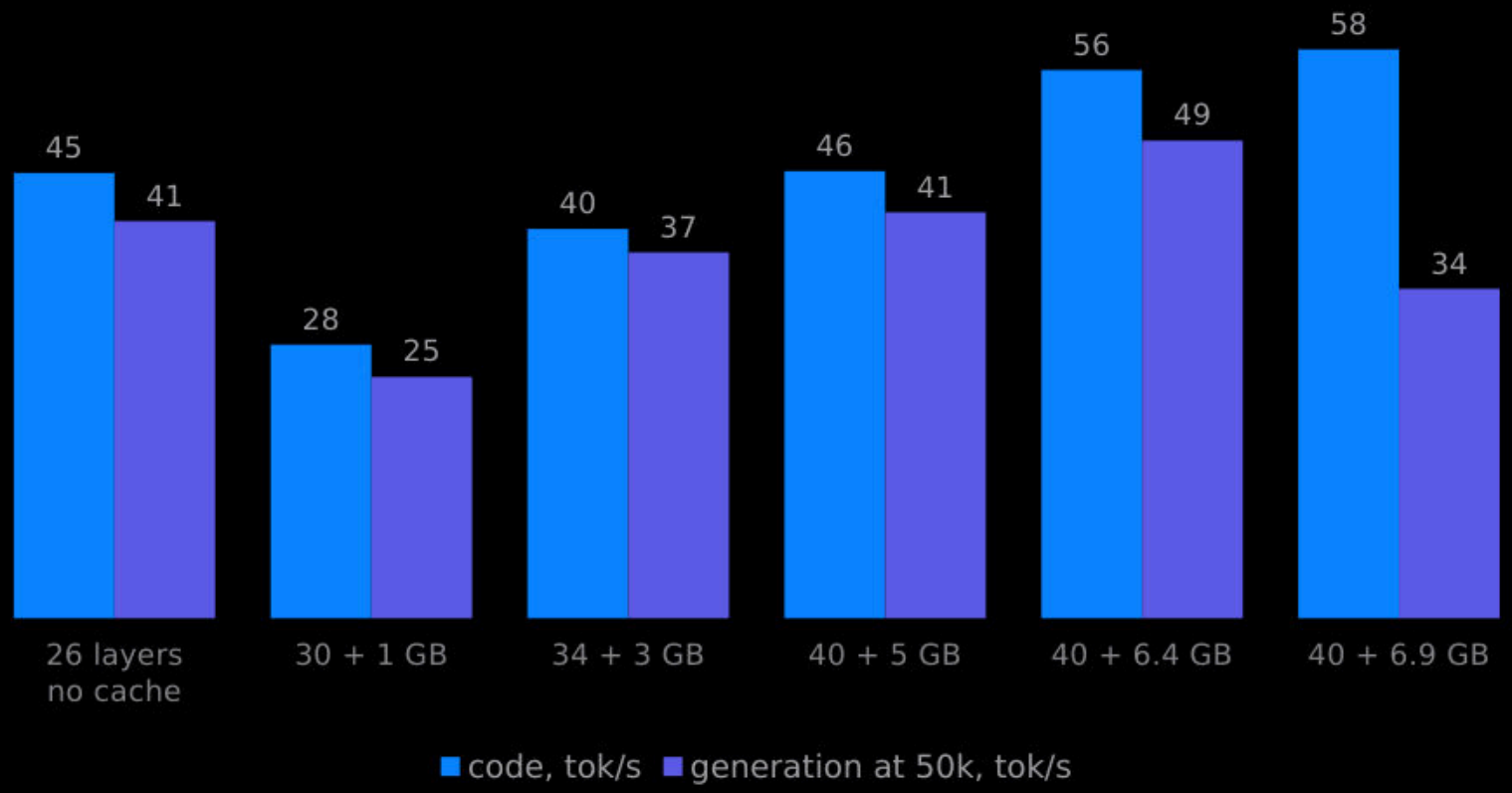
code · at 50k: 40 → 49

- new in llama.cpp v0.7.0
- 6,900 MiB overflows at 50k

Tried and dropped: KAT-Coder-V2.5 (41.9 tok/s, worse on agent tasks) · thinking mode (24 tok/s, ~10k tokens per task) · Kaggle TPU (platform bug)

Tuning

Six runs, one winner



40 + 6.4 GB

the winner

All experts in RAM + an LRU cache on the GPU beats the old "26 layers on the card".

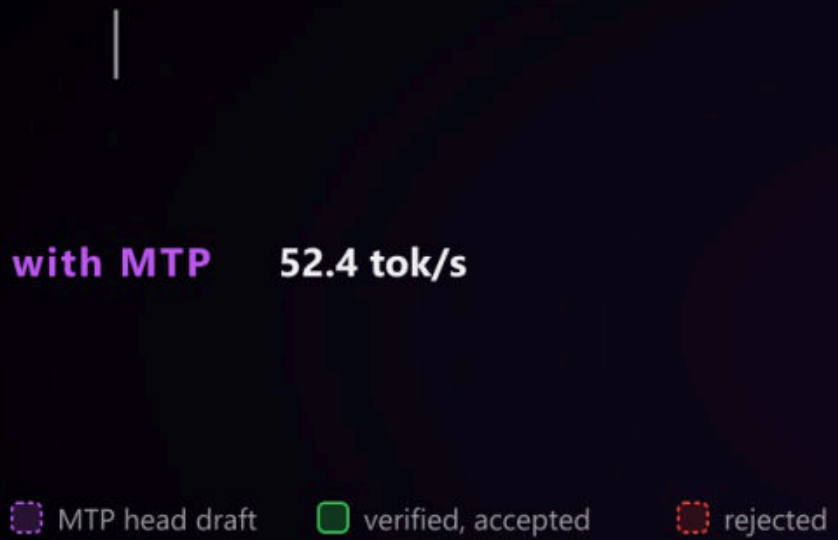
6.9 GB is faster on short prompts, but at 50k context VRAM overflows: 48.7 → 33.6 tok/s.

Multi-token prediction

+44 %

faster code generation.
The model drafts two tokens ahead.

spec-type = draft-mtp · same model, same GPU



models.ini

The parameters, and why these

n-cpu-moe = 40	all experts to RAM, the GPU stays for attention
moe-cache-mib = 6400	expert cache on the GPU; 6900 already floods VRAM
ctx-size = 65536	64k context; the KV cache is small on a hybrid (680 MB)
cache-type-k/v = q8_0	half-size KV cache, same quality, 2× faster than f16
batch 4096 / ubatch 2048	experts cross PCIe once per 2048 tokens: 2× prefill
spec-type = draft-mtp	MTP head, 2 tokens ahead: +44 % on code
reasoning = off	Qwen ignores reasoning-budget, this turns it off
--models-max 3	Qwen on the GPU, d1 and LFM3-VL on the CPU, all at once

Benchmarks

How fast it runs

~70
tokens/s generating text

805
tokens/s prefill on a 27k prompt

64k
tokens of context

11.6
GB of 12 GB VRAM in use

“Faster than you can read. Not that you read fast.” — GLaDOS

03

Brain and agent

What I use Qwen 3.6 for and how pi works.



Base model

What I use Qwen 3.6 for

Chat with a personality

the GLaDOS persona from SOUL.md

Web and research

searches, opens, summarizes

Coding

#coding right in my repos

Writing news

CSS news in Czech

Prompt translation

Slovak → English for images

Nothing leaves.

The model is local, so it may read private things too: files, the screen, email.

Agent

pi: an agent as a library

```
const {session }=await
  createAgentSession({
    model: qwen,       // local 35B
    resourceLoader,    // SOUL.md +skills
    customTools: [progressUpdate],
  });
await session.prompt(message);
```

Tools

bash, files, web_search, browse, codemode

Skills = markdown

free games, agent-browser, agent-desktop

d1 classifier

the llama.cpp provider sees it natively

pi 1.5.0

since 2 Nov, nightly update with 11 tests

04

Discord and decisions

The bridge, slash commands and a small model that only decides.



Interface

Discord bridge

1,900

lines of JavaScript between Discord and the agent

0

Channel = conversation

survives restarts

No mentions needed

across my whole server

Only me

allow-list, the agent has a shell

Progress updates

screenshots included

A model per channel

local or free from OpenRouter

Tokens out of the shell

the agent can neither see nor print them

Commands

18 slash commands, mostly plain

CODING & AGENT

/ask	ask GLaDOS anything
/search	web search, no guessing
/browse	a page in a real browser
/skill	run one of the skills
/model	show / switch the model
/switch-free-coding-models	free OpenRouter coder
/get-new-models	new stealth + free models
/status	model, router, context
/reset	fresh conversation

PERSONAL

/get-news	the AI news brief, now
/check-games	free games, optional claim
/key	redeem a Steam key
/what-the-track	Rádio FM now + Spotify link
/add-to-playlist	track → the right playlist
/cinema	Cinema City listings for a day

CSS CZECHOSLOVAKIA · PUBLIC SERVER, OWNER ONLY

/css-news	Chrome + WebKit news
/css-ai-news	AI in CSS and frontend
/css-tips-and-tricks	CSS tips and tricks

Decision model

d1-2B: never writes, only decides

STATE

headline: “Genie 4 …”

QUESTION

Is this real AI news?

ANSWER

yes · 87 %

1 GB

on the CPU next to Qwen · /v1/systemone endpoint

TEST ON MY OWN DATA	d1-3B	d1-2B
news: passed / noise (of 8 / 58)	4 / 4	**5 / 0**
duplicate headlines (of 6)	4	**6**
time per question	6.4 s	**4.0 s**
DnB tracks (12) / false hits (24)	11 / 1	**10 / 0**

/add-to-playlist

• LIVE DEMO

Code for facts. d1 for judgment.

1 · code

what Rádio FM plays, find it on Spotify

2 · code

MusicBrainz: is the artist CZ/SK?

3 · d1

Is it drum and bass? Is it indie?

4 · code

add to DnB [2026], suggest more

10 / 12

DnB tracks recognized

0 / 24

other genres let through

News filter

1,400 headlines, three items

AI news · 31 Oct · llama.cpp

kv-cache: keep rotation constants on the GPU
d1: cuda 45 %

ggml: add batched tensor copy API
d1: feature 95 %

common: drop the Windows argv reconstruction
always watched: Windows

Every hour

RSS, sites without RSS, GitHub

P ≥ 0.85

plus a d1 duplicate check

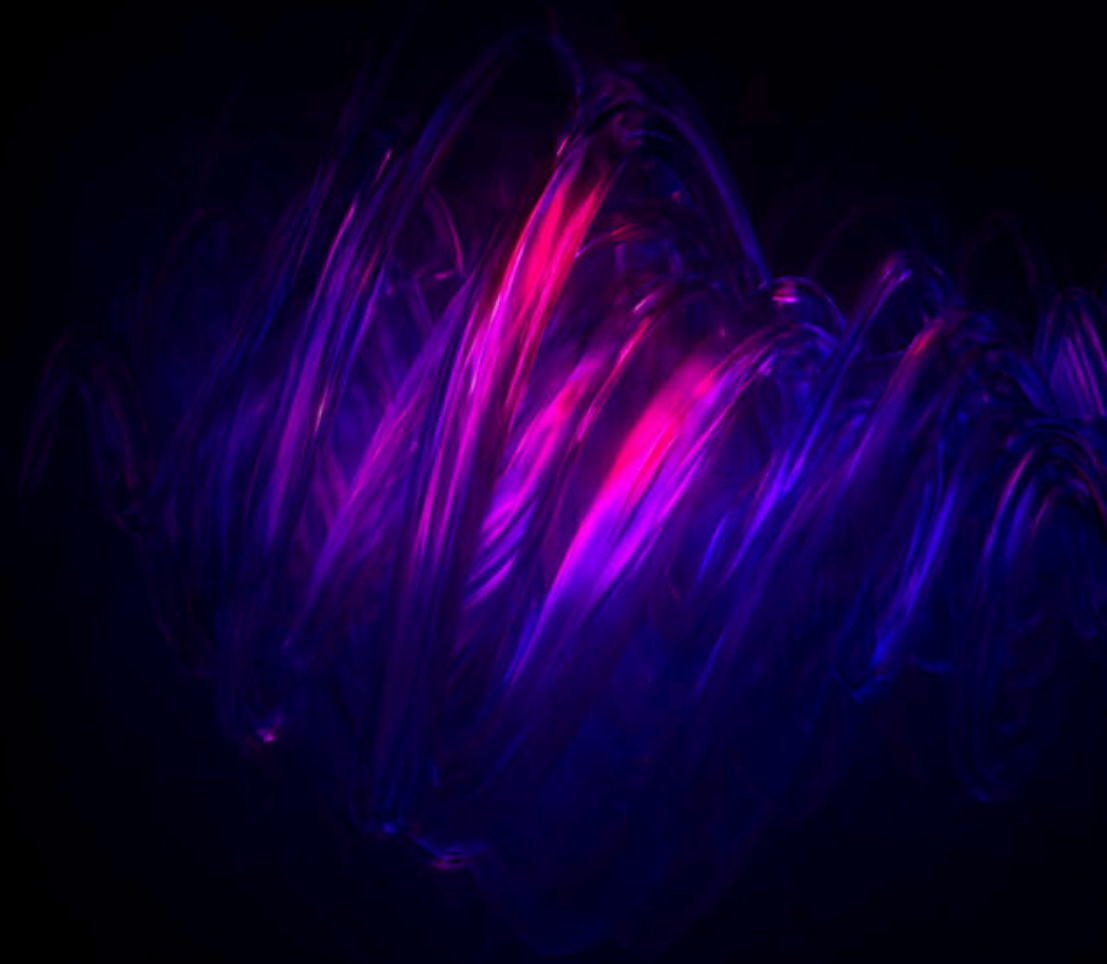
CSS community

#novinky, #ai, #tipy-a-triky, #udalosti

05

Voice, vision and hands

Voice notes, images, the browser and the whole desktop.

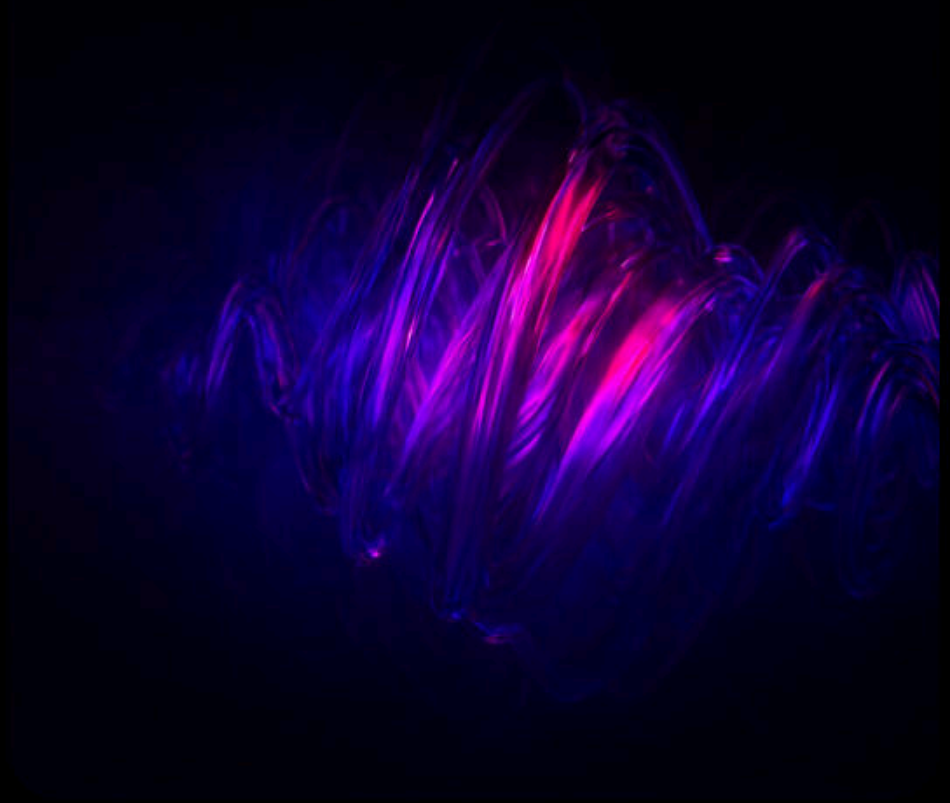


Voice

• LIVE DEMO

Talk instead of typing

A voice note from the phone → Whistle transcribes it → GLaDOS replies. In any channel.



0.3 s

to transcribe 10 s of audio on the CPU

17 MB

the whole Whistle model

No Slovak: “Slovenský jazyk” → “Somenski y Azik.”

Vision

• LIVE DEMO

Qwen is blind, so it got eyes

NOW		TODAY		HISTORY		TOKENS	
89 W whole PC (est.)		**1.61 kWh**		yesterday	1.71 kWh · CZK 10.28	**5,176** tokens today	
GPU	15 W · 42 °C · 4 %	PC on	22.7 h	last 7 days	11.69 kWh · CZK 70.14	LLM busy today	10 min · prefill 74,410
CPU + RAM	29 W · 24 %	while the LLM generated	21 Wh · 10 min	this month	3.33 kWh · CZK 19.96	energy per 1k tokens	7.14 Wh
VRAM	11.3 / 12.0 GB	extra over idle	10 Wh (idle ≈ 67 W)	LLM this month	35 Wh · CZK 0.21 · 16 min	energy per 1M tokens	7.14 kWh
		cost	CZK 9.68 · LLM CZK 0.13			cost per 1M tokens	CZK 42.82

READ BY LFM3-VL-2B · 18.8 s · CPU ONLY

NOW: 89 W whole PC · GPU 15 W · 42 °C · 4 %
TODAY: 1.61 kWh · while the LLM generated 21 Wh · 10 min
HISTORY: last 7 days 11.69 kWh · CZK 70.14
TOKENS: 5,176 tokens today · cost per 1M tokens CZK 42.82

0 GB

VRAM

100%

of the numbers read right

#image-generator

• LIVE DEMO

Draws and edits: Qwen-Image 2.1



„DoLorain flying above High Tatras on Slovakia“

reply in the chat

„Don’t see flying car above Tatras“

→



car added, top centre

~70 s quality model **~20 s** Turbo (“rýchlo …” = fast)

Model swap

One GPU, two big models

1 Qwen translates the prompt

2 router: unload Qwen

3 sd.cpp draws

4 router: load Qwen

VRAM DURING ONE IMAGE · MEASURED

Qwen 35B · 11.5 GB

Qwen-Image · ~6 GB, GPU 100 %

Qwen back · ~20 s

Other jobs wait on a GPU lock, so the router never loads Qwen back into a full card.

agent-browser

• LIVE DEMO

A real browser for the agent

```
#in the #coding channel
agent-browser open localhost:5173
agent-browser snapshot   #buttons with refs
agent-browser click @e12
agent-browser console
agent-browser screenshot shot.png
```

Tests what it just built

dev server, click, console, screenshot

Text, not pixels

the page as an element tree with refs

Screenshot to the chat

proof via progress_update

agent-desktop

• LIVE DEMO

It drives Windows apps too

```
agent-desktop launch notepad
agent-desktop snapshot --app Notepad -i
agent-desktop set-value @s3hf…:e1 "Hello"
agent-desktop screenshot --app Notepad
{"ok":true,"command":"launch"}
```

Accessibility tree

like agent-browser, for windows

Leaves the mouse alone

I can keep working

Local model only

the screen never goes to the cloud

06

What broke

And what it all costs.



Incidents

Four bugs, four lessons

Armoury Crate ate the memory

handle leak → commit limit → llama.cpp crashed with 0xc0000409

Qwen kept thinking anyway

it ignores reasoning-budget = 0; only reasoning = off worked

The monitor woke the models

WattLlama reloaded an unloaded model within 150 ms

Forgotten dev servers

5 servers ran for weeks, 4 of them on 0.0.0.0

“Bugs are not failures. They are data. Sometimes rather expensive data.” — GLaDOS

WattLlama

What it costs

Measured 27 Oct – 2 Nov, electricity at CZK 6/kWh

CZK 0.97

per million tokens above idle

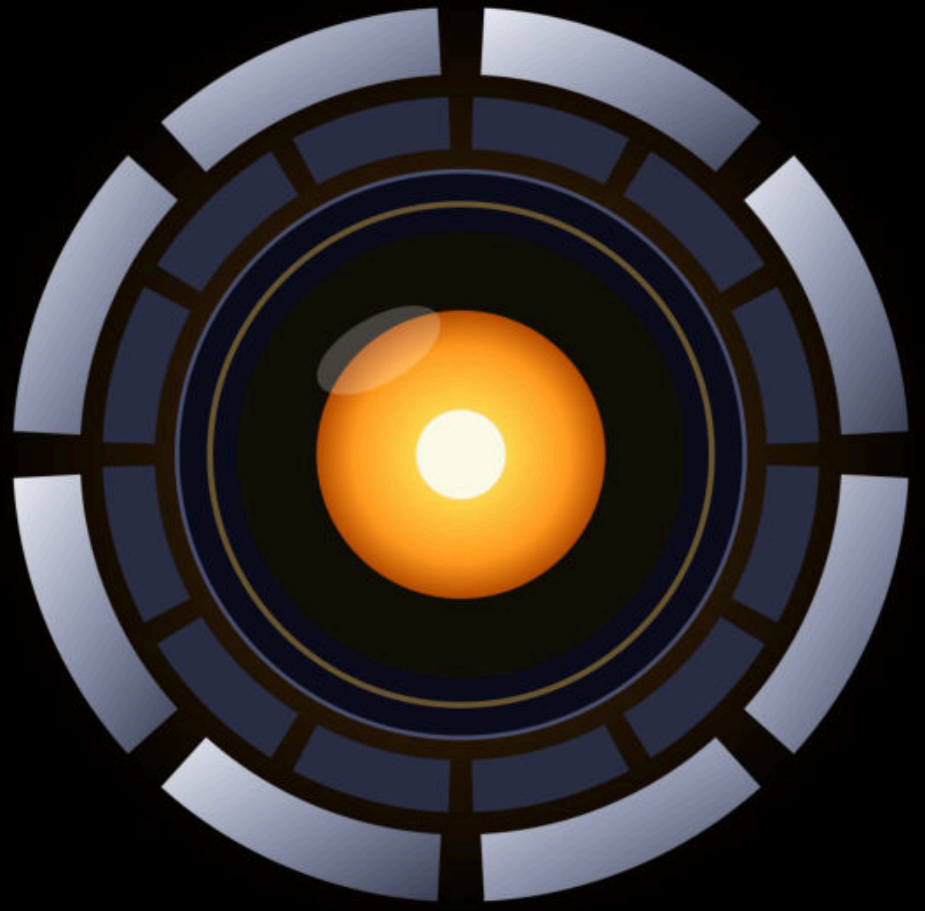
CZK 70

the whole PC 24/7 · 11.7 kWh · 67 W idle



Test complete.

Thank you



Lukas Chylik · github.com/depthark

pi coding agent · llama.cpp · stable-diffusion.cpp
Qwen 3.6 · Qwen-Image 2.1 · LiquidAI d1 and LFM3-VL
Cactus Whistle · agent-browser and agent-desktop

WattLlama
github.com/depthark/wattllama